KasSigner Quick Start Guide

Air-gapped Kaspa signing - setup, recovery, review, and secure ownership

Security notice. Verify addresses, amounts, fees, change, wallet identity, and warnings on the hardware screen before approving. KasSigner uses consumer ESP32-S3 hardware and is not a secure-element or tamper-resistant device. Recovery words and any BIP39 passphrase remain the durable cross-device backup.

1. What you need

Part	Requirement
Hardware	M5Stack CoreS3 is hardware-tested. Waveshare-family ESP32-S3 targets are source-supported and require separate hardware qualification.
Companion	KasSee Web, Android, or iOS. Mobile apps host the same KasSee Rust/WASM runtime.
Recovery	12/24 BIP39 words plus optional BIP39 passphrase. Do not rely on device-bound ciphertext as your only recovery path.
Host	Linux or Windows for the complete native build/QA bootstrap; macOS/Xcode for iOS development.

2. Build or run the companion

From the repository root: `make kassee`, `make android`, or `make ios`. For firmware development use `make firmware`. Flashing is always explicit with `make flash BOARD=... PORT=...`.

3. Create or restore a wallet

- Create a new 12/24-word wallet or open Wallet -> Switch / Add Wallet -> Restore Wallet.
- Restore Wallet first asks for a source: Words / SeedQR / SD / Advanced.
- Mnemonic restore order: source -> optional BIP39 passphrase -> wallet name -> Save Securely / Session Only. Imported words are not shown back after entry.
- Account XPrv or raw-key restore skips the BIP39-passphrase step; wallet naming and Save/Session choice still apply.
- Session Only is RAM-only and is excluded from persistent wallet serialization.

4. Back up and recover

- Wallet -> Backup contains common methods; Backup -> Advanced contains Compact/plain SeedQR, steganographic backup, XPrv backup, and raw-key export where applicable.
- Wallet -> Recovery contains import/recovery methods. Backup and Recovery are direct Wallet entries; there is no intermediate Backup/Recovery screen.
- Device-bound encrypted backups require the creating device. Portable disaster recovery is the mnemonic plus optional passphrase; Portable steganographic backup is JPEG + strong password.

5. Pair KasSee and sign

- Export the wallet kpub/watch-only identity from KasSigner and import it into KasSee.
- Create the transaction in KasSee. Confirm recipient, amount, fee, change, and network, then display the PSKT/PSKB or KSPT v4 signing QR.
- Scan the complete request with KasSigner. Review the hardware summary; descriptor-proven change must not be counted as recipient spend.
- Approve only if the hardware view is correct. Scan the signed response back into KasSee and broadcast.

6. Optional CoreS3 owner authority

On production CoreS3 hardware, **Pop It!** permanently enables ESP32-S3 Secure Boot v2. If you want an independent owner application-signing key, enroll it first at **Settings -> Advanced -> Owner Firmware -> Enroll Owner Key**. Pop It! performed first permanently closes owner enrollment.

Build enrollment/install media with `make owner-firmware OWNER_KEY=/secure/offline/path/owner.pem`. `OWNERKEY.KAS` contains only the owner public-key digest and checksum; `OWNERFW.BIN` is the owner-signed application.

> **Irreversible trust warning:** owner enrollment preserves the vendor and owner authorities and write-protects their revoke controls. A lost owner key does not block official updates, but a compromised enrolled key remains trusted by that device. Keep the RSA-3072 private key offline and backed up.

Development firmware validates/simulates this workflow only. It cannot arm the production boot-control/eFuse transition.